UNIT 4: Statistical Data

4.1: No-Code AI for Statistical Data

Lesson Title: No-Code AI for Statistical Data	**Approach:** Session + Activity
Summary: Understanding the importance and need of No-Code AI for Statistical Data, advantages, and disadvantages of using No-Code AI for Statistical Data, how to use No-Code AI tools for Statistical Data.	
Learning Objectives: • Understand the meaning of No-Code AI and the need of it. • Understand the difference between No-Code and Low-Code. • What are some no-code tools used for the statistical dataset? • How is the AI project cycle different for No-Code AI?	
Learning Outcomes: • Define No-Code and Low-Code AI. • Identify the differences between Code and No-Code AI concerning Statistical Data. • Relate AI project stages to the stages of No-Code AI projects.	
Pre-requisites: Basic knowledge of AI project cycle.	
Key-concepts: • No code tools for AI projects • Understanding different types of approaches like code, low code and No-Code for statistical data. • Identifying and recognizing various tools and platforms for low code and No-Code approaches	

Data science is a journey of exploration and discovery. Artificial Intelligence is a technology which completely depends on data, which is fed into the machine which makes it intelligent. And depending upon the type of data we have; AI can be classified into three broad domains: Data science, Computer Vision and Natural language processing.

Data Sciences is a concept to unify statistics, data analysis, machine learning and their related methods in order to understand and analyze actual phenomena with data. It employs techniques andtheories drawn from many fields within the context of Mathematics, Statistics, Computer Science, and Information Science.

What is Data Science ?

Applications of Data Science:

Internet Search: All the search engines (including Google) make use of data science algorithms to deliver the best result for our searched query in the fraction of a second. Considering the fact that Google processes more than 20 petabytes of data every day, had there been no data science, Google wouldn't have been the 'Google' we know today.

Targeted Advertising: If you thought Search would have been the biggest of all data science applications, here is a challenger – the entire digital marketing spectrum. Starting from the display banners on various websites to the digital billboards at the airports – almost all of them are decided by using data science algorithms. This is the reason why digital ads have been able to get a much higher CTR (Call-Through Rate) than traditional advertisements. They can be targeted based on a user's past behaviour.

Website Recommendations: Aren't we all used to the suggestions about similar products on Amazon? They not only help us find relevant products from billions of products available with thembut also add a lot to the user experience. A lot of companies have fervidly used this engine to promote their products in accordance with the user's interest and relevance of information. Internet giants like Amazon, Twitter, Google Play, Netflix, LinkedIn, IMDB and many more use this system to improve the user experience. The recommendations are made based on previous search results for auser.

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Genetics & Genomics:

Data Science applications also enable an advanced level of treatment personalization through research in genetics and genomics. Data science techniques allow integration of different kinds of data with genomic data in disease research, which provides a deeper understanding of genetic issues in reactions to particular drugs and diseases. As soon as we acquire reliable personal genome data, we will achieve a deeper understanding of human DNA. The advanced genetic risk prediction will be a major step towards more individual care.

Activity: Word Scramble the terms related to AI applications.

Purpose: Recall of AI terms

VANAGTOINI APP ______________________________

UALTIRV SSISATANT ____________________________

AGEGUALAN TIONSLATRAN ________________________

Introduction to Low/No-Code AI approach for Statistical Data

Let's say you want to build a product, food delivery application. How do you go about starting it?

Building a food delivery application involves several steps, from conceptualization to development, testing, and deployment. The 3 most popular approaches to code are given below.

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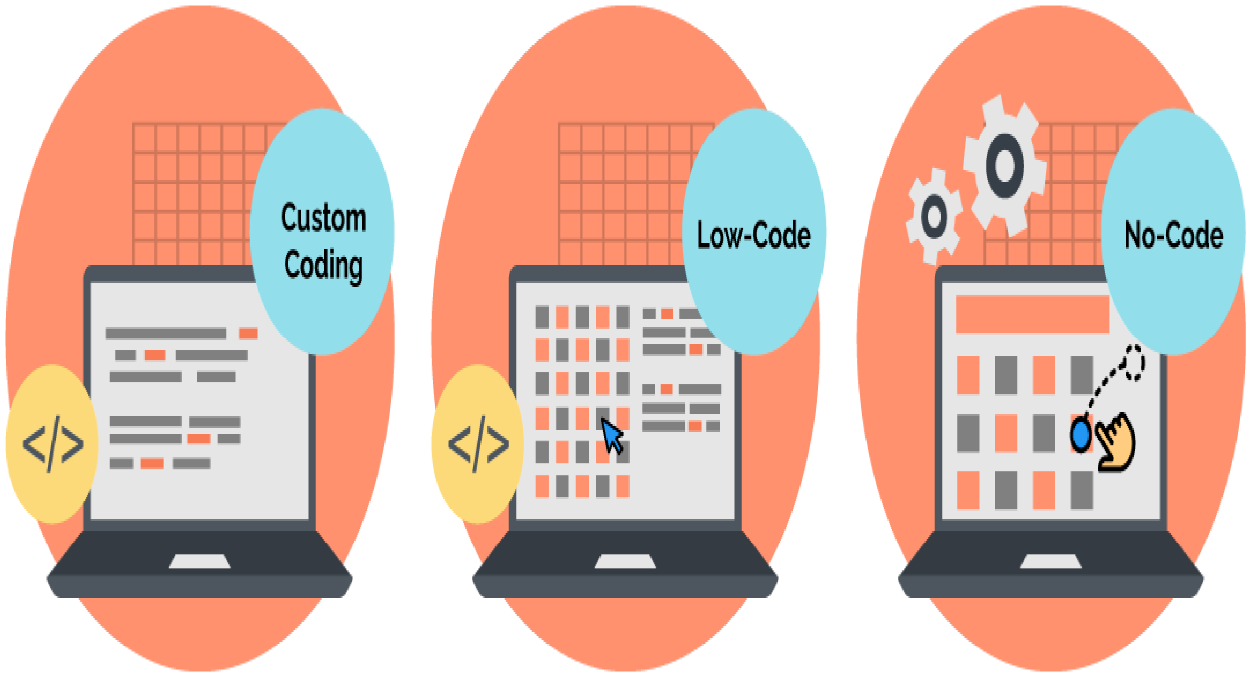

Custom code is also known as high code.

How do we choose? Which of these 3 is the most suitable for our app?

High code	Low code	No code
High code development refers to traditional software development where programmers write code manually using programming languages like Java, Python, C#, etc.	Low code development involves using platforms or tools that provide visual interfaces and pre-built components to streamline the application development process.	No code development takes low code principles further by allowing users to create applications without any coding or scripting knowledge.
A team of software coders need to write all the code manually.	Programmers need to write some code manually.	Coding knowledge is not required; hence anyone can make the product.
It is expensive.	It is less expensive compared to high code.	It is less expensive compared low code.
The company can own the product they create. You can create anything and customise your product in any way.	You can customise your product to an extent only using code. For example, custom chatbot.	Lack of customisable options as No-Code AI tools are limited to functions in the tool. Simple to use as it uses drag-and-drop features instead of coding.

Now that we have seen the differences,
which approach do you think is the most suitable one for our Food Delivery app? Discuss!

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Can you think of an invention that has made life easier in terms of saving time/cost for you?

Some inventions that have made life today easier are smartphones, credit cards, internet, onlinestreaming services, Refrigeration technology, GPS navigation, medical innovations etc.

Similar to those inventions, let's look at how No-Code AI makes our lives easier!

```
models = []
models.append(('LR', LogisticRegression(solver='liblinear', multi_class='ovr')))
models.append(('LDA', LinearDiscriminantAnalysis()))
models.append(('KNN', KNeighborsClassifier()))
models.append(('CART', DecisionTreeClassifier()))
models.append(('NB', GaussianNB()))
models.append(('SVM', SVC(gamma='auto')))
# evaluate each model in turn
results = []
names = []
for name, model in models:
    kfold = StratifiedKFold(n_splits=10, random_state=1, shuffle=True)
    cv_results = cross_val_score(model, X_train, Y_train, cv=kfold, scoring='accuracy')
    results.append(cv_results)
    names.append(name)
    print('%s: %f (%f)' % (name, cv_results.mean(), cv_results.std()))
```

```
# Make predictions
from pandas import read_csv
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import accuracy_score
from sklearn.svm import SVC
# Load dataset
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/iris.csv"
names = ['sepal-length', 'sepal-width', 'petal-length', 'petal-width', 'class']
dataset = read_csv(url, names=names)
# Split-out validation dataset
array = dataset.values
X = array[:,0:4]
y = array[:,4]
X_train, X_validation, Y_train, Y_validation = train_test_split(X, y, test_size=0.20, random_stat
# Make predictions on validation dataset
model = SVC(gamma='auto')
model.fit(X_train, Y_train)
predictions = model.predict(X_validation)
# Evaluate predictions
print(accuracy_score(Y_validation, predictions))
print(confusion_matrix(Y_validation, predictions))
print(classification_report(Y_validation, predictions))
```

More code to test out different algorithms…And

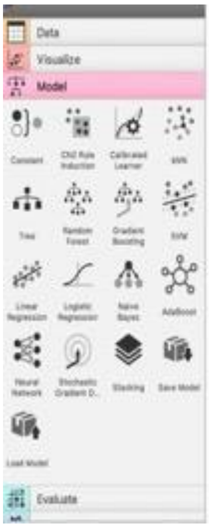
more code to pick the best algorithm…

```
# evaluate each model in turn
results = []
names = []
for name, model in models:
    kfold = StratifiedKFold(n_splits=10, random_state=1, shuffle=True)
    cv_results = cross_val_score(model, X_train, Y_train, cv=kfold, scoring='accuracy')
    results.append(cv_results)
    names.append(name)
    print('%s: %f (%f)' % (name, cv_results.mean(), cv_results.std()))
# Compare Algorithms
pyplot.boxplot(results, labels=names)
pyplot.title('Algorithm Comparison')
pyplot.show()
```

That's a lot of code, right? And that's why we have No-Code AI.

No-Code

- In No-Code AI, we can drag and drop, these models in few seconds.
- No coding knowledge is required to implement complex ML algorithms
- Drag and drop feature of a No-Code tool makes it easier.



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Why do we need No-Code AI?

- We tend to run into many types of errors when we are coding, and it can be very troublesome at times.
- In No-Code AI since we do not need to code, we won't have any code errors!
- No-Code AI helps to save cost for businesses as it is costly to implement completely coded AI systems.
- Companies can implement AI with less stress and without the need to hire an AI staff with No-Code AI.
- No-Code AI is easy to use – even middle school students can create AI using No-Code tools
- Since it has visual & drag-and-drop features, anyone can see what they are building in real-time

Who can use No-Code AI?

- No-Code AI makes AI more accessible to the general public.
- Non-technical people such as doctors, architects, musicians may quickly construct accurate AI models with no coding involved.

Let's look at a scenario to understand who can use No-Code AI

- No-Code AI makes AI more accessible to the general public.
- Non-technical people such as doctors, architects, musicians may quickly construct accurate AI models with no coding involved.

Thus No-Code AI can empower individuals and organizations across various industries and skill levels to harness the potential of artificial intelligence for their specific needs.

Let's look at a scenario to understand who can use No-Code AI.

Problem: Kayla is a wildlife animal's dietitian manager at the zoo. She takes care of the cost of buying meat and vegetables for animals. With the prices of food increasing rapidly, it will become more expensive for the zoo to buy healthy and nutritious foods for its animals. Therefore, the zoo's accounts team wants to know the increase in the price of food so that they can ask the government or sponsors to fund for the food. Thus, Kayla requires the help of AI to predict the price.

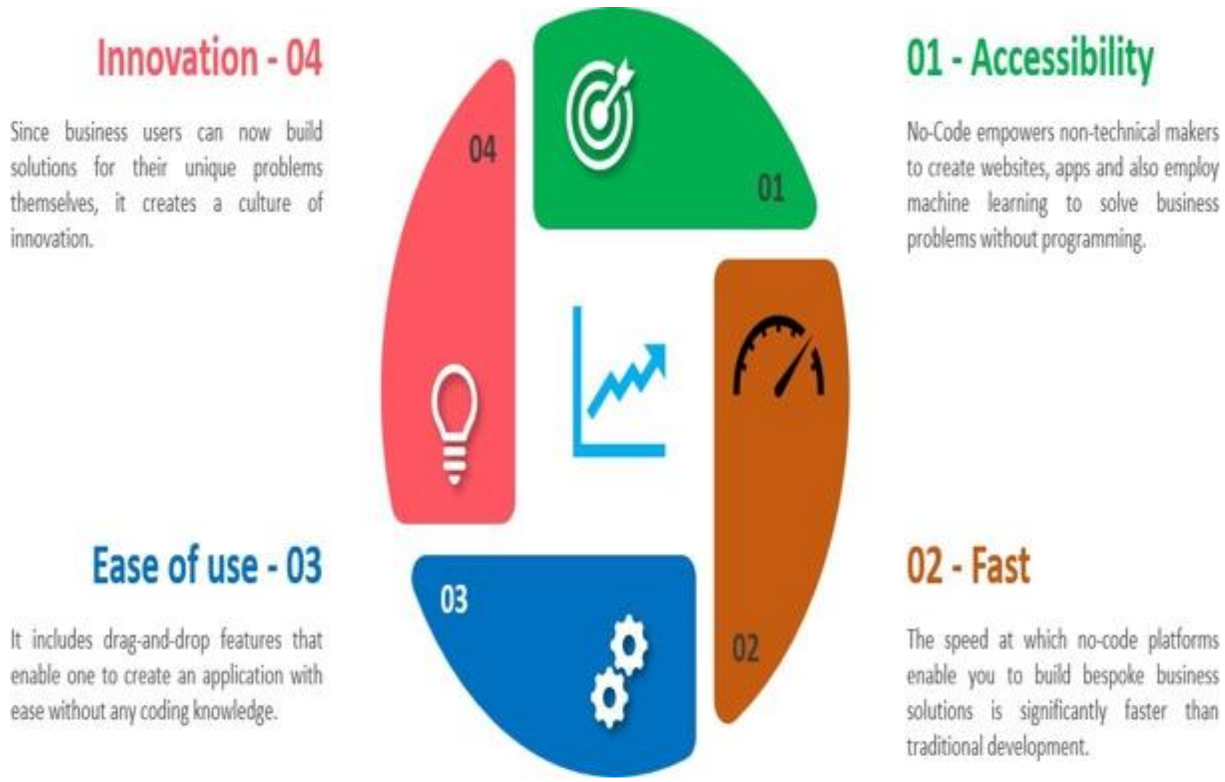
However, Kayla does not know how to code to build an AI model as she has not learnt AI before.

Solution:

She finds a No-Code AI tool online that can help her build an AI model to predict the prices of meatand vegetables without any code!

Kayla can also use any popular No-Code Tool like Orange Data Mining to build a price prediction model. We will look at how to build a simple price prediction model in a short while. From this scenario you can infer that: You do not need to know coding to build an AI model and anyone canmake use of No-Code AI to build an AI model.

Benefits of No-Code Tools



Now that we have discussed the advantages of No-Code tools, let us discuss the disadvantages too…

Disadvantages of No-Code Tools

Lack of Flexibility

Drag-and-drop elements can be very convenient. On the other hand, you are limited to those fixed elements. Hence, you have a limitation in customizing your application in no-code platforms, and it does not give you all the authority and flexibility you might need.

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Automation Bias

Automation bias is the tendency for humans to favor suggestions from automated decision-making systems and to ignore contradictory information made without automation, even if it is correct.

Security Issues

No code platforms do not essentially force you to think of security first or even evaluate security best practices. Therefore, these applications are only best suited for companies that don't deal with sensitive data, as these platforms usually offer limited control over it.

Popular No-Code Tools

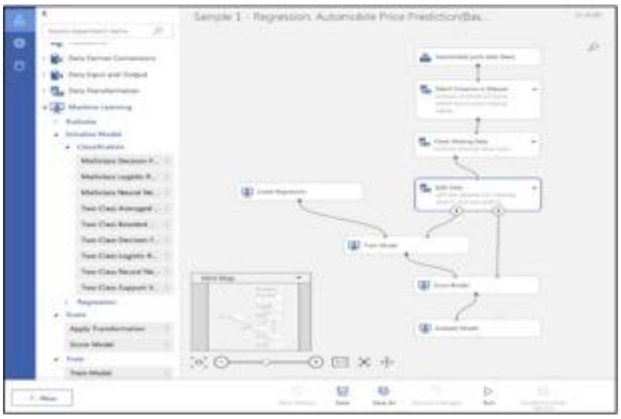
Here are some of the top No-Code AI tools where we can build our AI models in.

No-Code tool	Details	Released
Azure Machine Learning	Cloud-based service provided by Microsoft	July 2014
Google Cloud AutoML	Cloud-based service provided by Google	January 2018
Orange Data Mining	An open-source data visualization, machine learning and data mining toolkit. Developer: University of Ljubljana	October 1996
Lobe AI	Lobe AI is a machine learning platform that enables to create custom machine learning models using a visual interface	2015
Teachable Machine	Teachable Machine is a web-based tool that makes creating machine learning models fast, easy, and accessible to everyone.	November 2017

Azure Machine Learning

Microsoft's Azure Machine Learning aims to simplify ML processes.

- It allows users to build ML models without having to touch any code,
- and provides an easy interface for cleaning up your data, training models, evaluating them and, finally, putting them into production.



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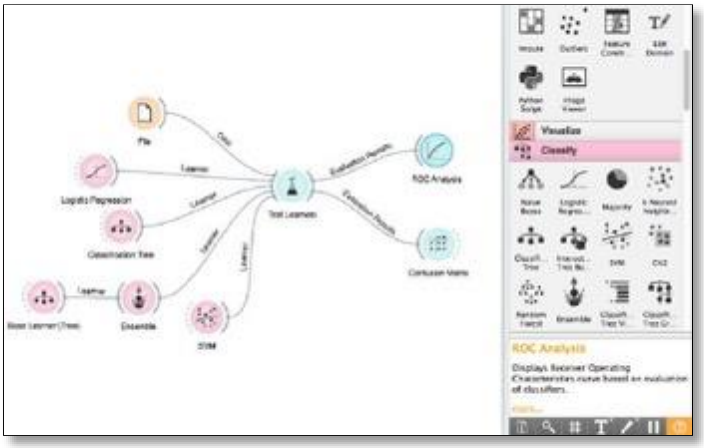
Google Cloud AutoML

- Users with limited ML knowledge can train high-quality models specific to their business needs with minimal effort.
- They can build their own custom machine learning model in minutes and then use the models in their applications and web sites.

  Click on Introducing Cloud AutoML to watch a video on it.

Orange Data Mining



- Orange enables you to visualize data and perform data mining and machine learning.
- You can use this tool without writing a single code.
- This platform can be used for analysis, is relatively easy, and has beautiful visuals.

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Now, let us get back to building a simple prediction model using Orange Data Mining for Kayla!

FAO food price index

		Food Price Index [1]	Meat [2]	Dairy [3]	Cereals [4]	Vegetables Oils [5]	Sugar [6]
2004		65.6	67.6	69.8	64.0	69.6	44.3
2005		67.4	71.8	77.2	60.8	64.4	61.2
2006		72.6	70.5	73.1	71.2	70.5	91.4
2007		94.3	76.9	122.4	100.9	107.3	62.4
2008		117.5	90.2	132.3	137.6	141.1	79.2
2009		91.7	81.2	91.4	97.2	94.4	112.2
2010		106.7	91.0	111.9	107.5	122.0	131.7
2011		131.9	105.3	129.9	142.2	156.5	160.9
2012		122.8	105.0	111.7	137.4	138.3	133.3
2013		120.1	106.2	140.9	129.1	119.5	109.5
2014		115.0	112.2	130.2	115.8	110.6	105.2
2015		93.0	96.7	87.1	95.9	89.9	83.2
2016		91.9	91.0	82.6	88.3	99.4	111.6
2017		98.0	97.7	108.0	91.0	101.9	99.1
2018		95.9	94.9	107.3	100.8	87.8	77.4
2019		95.1	100.0	102.8	96.6	83.2	78.6
2020		98.1	95.5	101.8	103.1	99.4	79.5
2021		125.7	107.7	119.1	131.2	164.9	109.3
2021	September	129.2	112.7	118.1	132.8	168.6	121.2
	October	133.2	112.0	121.5	137.1	184.8	119.1
	November	135.3	112.5	126.0	141.4	184.6	120.2
	December	133.7	111.0	129.0	140.5	178.5	116.4
2022	January	135.6	112.1	132.6	140.6	185.9	112.7
	February	141.2	113.9	141.5	145.3	201.7	110.5
	March	159.7	119.3	145.8	170.1	251.8	117.9
	April	158.4	121.9	146.7	169.7	237.5	121.5
	May	158.1	122.9	144.2	173.5	229.2	120.4
	June	154.7	125.9	150.2	166.3	211.8	117.3
	July	140.6	124.1	146.5	147.3	168.8	112.8
	August	137.9	122.0	143.4	145.6	163.3	110.5
	September	136.3	121.4	142.5	147.8	152.6	109.7

Purpose: To build an AI model to predict price using Orange Data mining AI tool

Steps for building a simple priceprediction model:

Step 1: Download the datasetfrom https://www.fao.org/worldfoodsituation/foodpricesindex/en/

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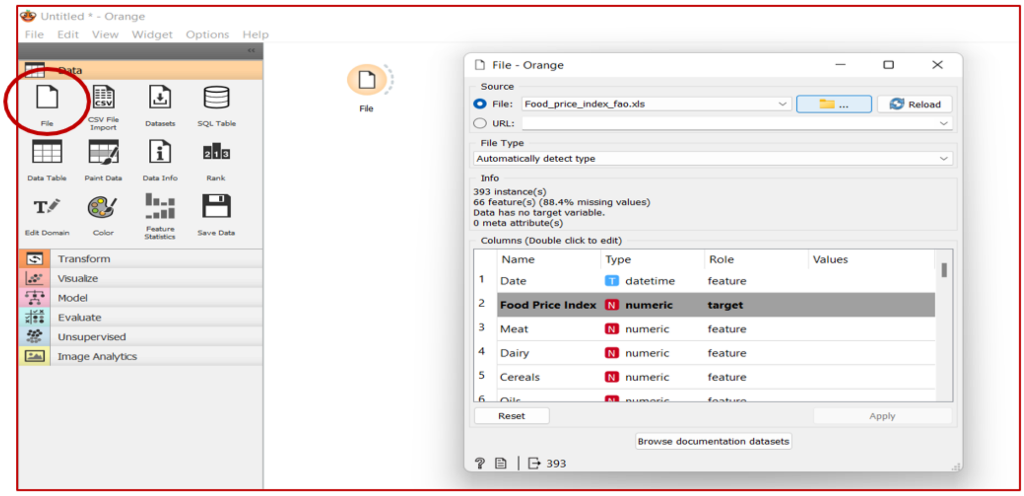
Step 2: Open Orange Data Mining

- Double Click on Orange Icon
- Open the tool

Step 3: Upload the Dataset

- Click on the File widget under Data Menu
- File widget will appear on the canvas
- Click on it and browse to the folder to upload the dataset
- Select the target variable as Food Price Index, as we are trying to predict the price index.



Step 4: View the Dataset

- Click on the Data Table widget under Data Menu
- Data Table widget will appear on the canvas
- Connect File to Data Table

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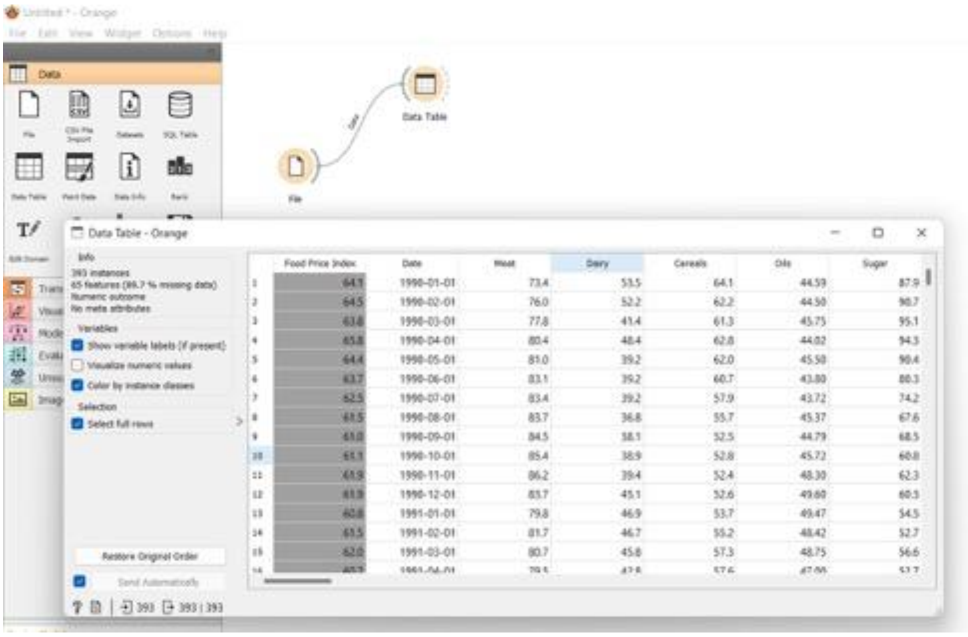
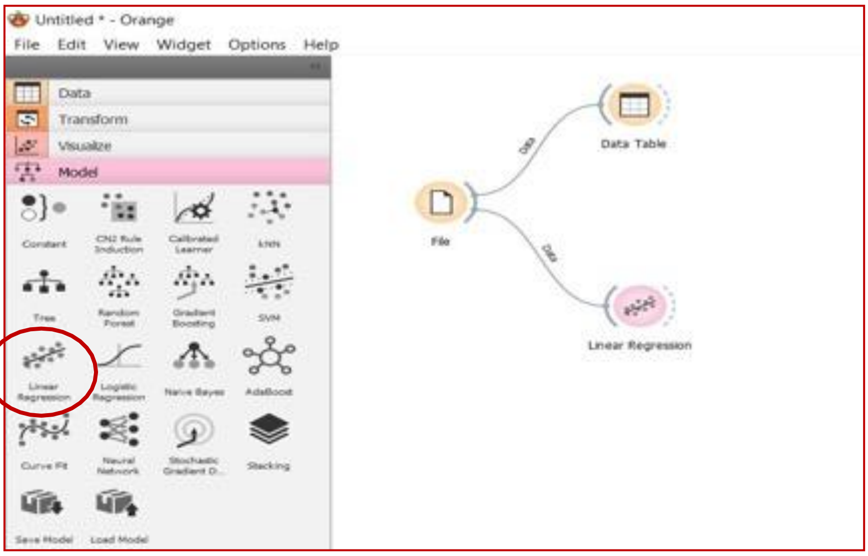
- Click on Data Table to view the dataset using the tool.



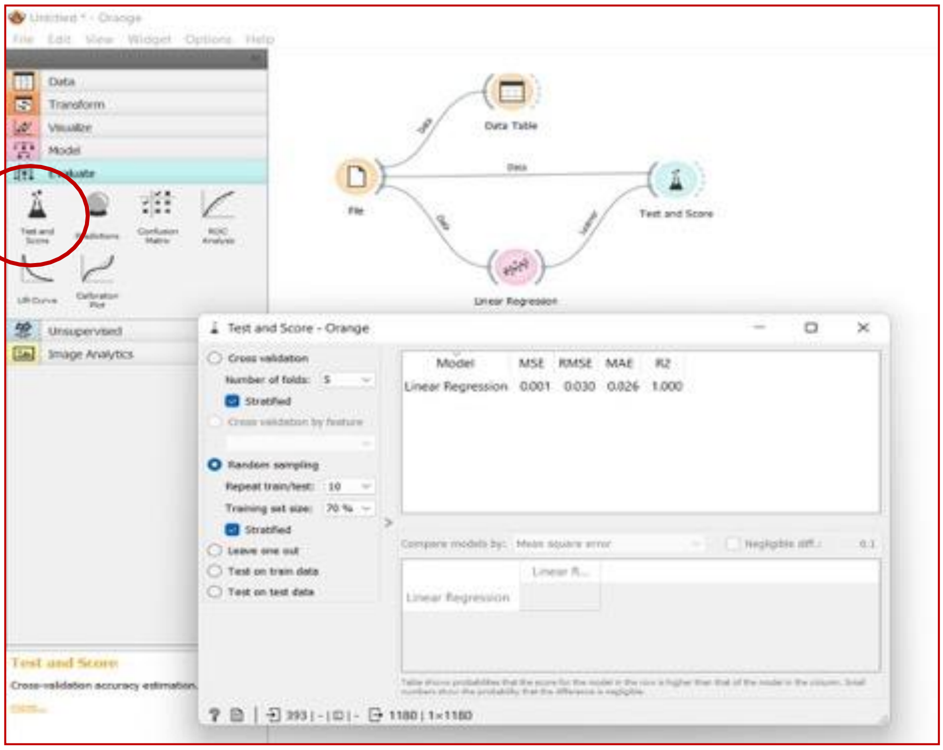
Step 5: Select the Model for Prediction

- Click on the Linear Regression widget under Model Menu
- Linear Regression widget will appear on the canvas
- Connect File to Linear Regression
- Linear Regression is an algorithm used for Regression Model



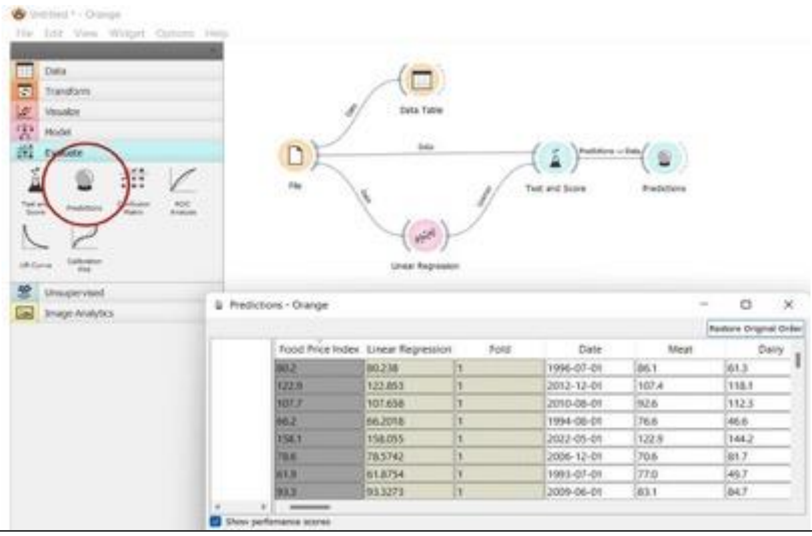
Step 6: Evaluate the Model

- Click on the Test and Score widget under Evaluate Menu
- Test and Score widget will appear on the canvas
- Connect File and Linear Regression to Test and Score to check the performance parameters.
- Click on Test and Score to view the parameters.



Step 7: Model Prediction

- Click on the Prediction widget under Evaluate Menu
- Prediction widget will appear on the canvas
- Connect the Test and Score to the Prediction to check the prediction made by Logistic Regression Model.
- Click on Prediction to view the price prediction.



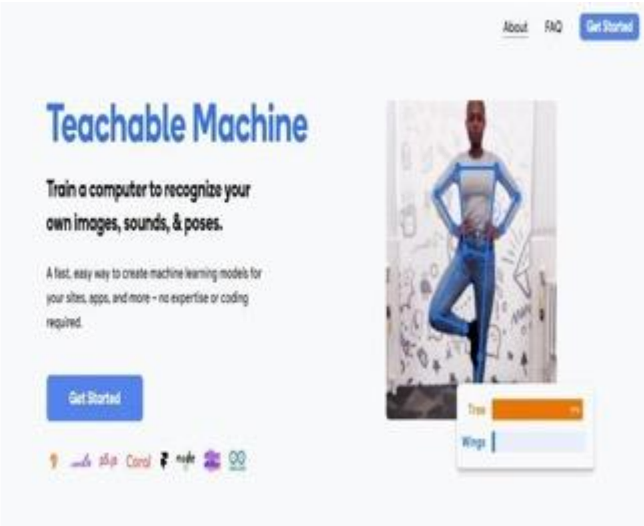
Kayla could use such a model to make the price prediction for animal feed and make a systematic fund-raising plan for the Zoo.

Other No code AI tools

Lobe

- Machine Learning has been made easy using Lobe.
- It has everything that you need to bring your machine-learning ideas to life.
- Lobe helps you, train models with a free and easy-to-use tool.
- It automatically trains a custom machine-learning model that can be shipped in your app.

Teachable Machine



- Teachable Machine is a web-based tool that makes creating machine learning models fast, easy and accessible to everyone.
- It can be used to train a computer to recognize your own images, sounds and poses.
- No specific expertise or coding is required to use this tool.

Test Yourself:

1. Orange data mining is an example of

a) Custom coding c) No code

b) Low code d) None of these.

2.Select Which is not the feature of No code approach.

a) visual c) code free

b) Highly expensive d) drag and drop

3. ____________ assembly relies on developers to write and deploy code.

a) High code c) No code

b) Low code d) None of these

c) 4.Flexibility is often limited in __________

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a) High code c) No code

b) High code and Low code d) High code and No code.

5. The organisation is heavily dependent on developer resources. This statement is true for

a) High code approach c) No code approach

b) Low code approach d) All of the above.

Answer the following briefly:

1. Name any two-cloud based No-code AI tools?
2. How accessible are no-code AI tools for non-technical users?
3. What types of tasks can be accomplished with no-code AI tools?
4. Can no-code AI tools be used for advanced projects? Justify.
5. Do no-code AI tools require prior programming knowledge? Justify.

Answer the following in detail:

1. What are the benefits of using No-code AI tools?
2. What are the challenges faced in using the No-code AI tools?
3. Differentiate low code and No-code AI tools with examples.
4. As the CEO of a small e-commerce startup, you're eager to leverage artificial intelligence to enhance your platform's user experience and drive sales. However, your team lacks the technical expertise to develop and deploy AI-powered solutions. What would be your recommendations for the CEO?
5. Samarth attended a seminar on Artificial Intelligence and has now been asked to write a report on his learnings from the seminar. Being a non-technical person, he understood that the AI enabled machine uses data of different formats in many of the daily based applicationsbut failed to sync it with the right terminologies and express the details. Help Samarth define Artificial Intelligence, list the three domains of AI and the data that is used in these domains.

Reflection Time:

1. Would you like to explore the tools specified here to build No-Code AI projects? How?

2. Are you aware of any other tools or platforms used for low code or No-Code AI Projects? List them.

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4.2: Statistical Data: Use Case Walkthrough

Lesson Title: Statistical Data: Use Case Walkthrough	**Approach:** Session + Activity

Summary: Introduction to a No-code AI tool for Statistical Data (Orange Data mining tool) through some practical use cases. Mapping AI Project cycle with use cases and perform data exploration, modelling and evaluation with Orange data mining.

Learning Objectives:

- Understand what is Orange Data Mining? What are its features?
- To perform Data Acquisition, Data Exploration, and Modelling on a No-Code tool.
- How to map the AI project cycle to develop an AI solution with a No-Code tool.
- Build the AI project using no code tool Orange Data Mining.

Learning Outcomes:

- Learner will be able to use no-code tool Orange Data mining.
- Learner will be able to map AI Project cycle with use cases.
- Learner will be able to perform data exploration, modelling and evaluation with Orange data mining.

Pre-requisites: Basic knowledge of AI project cycle.

Key-concepts:

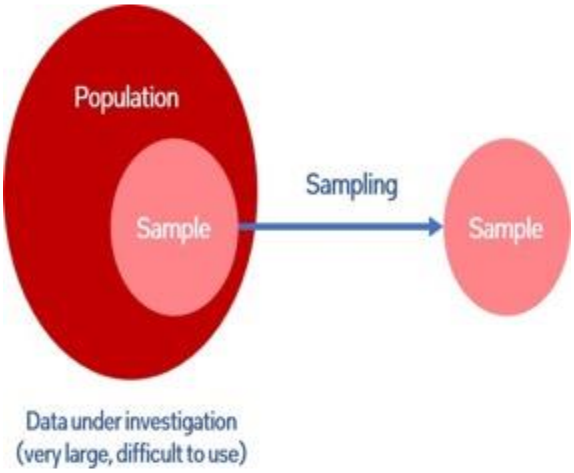
- Orange Data Mining is a machine learning tool for data analysis where we can perform operations through simple drag-and-drop steps.
- A sample of the entire population is taken to perform computations.
- Descriptive statistics – Mean, Median, Mode – help us to describe the data and its underlying characteristics.
- Distributions display the frequency of each value that appears in a data set.
- Probability is the likelihood of an event occurring.
- Variance measures how far each value in the data set is from the mean.

Before we start with Orange data mining, let's discuss some important concepts of Statistical Analysis

Important concepts in statistics

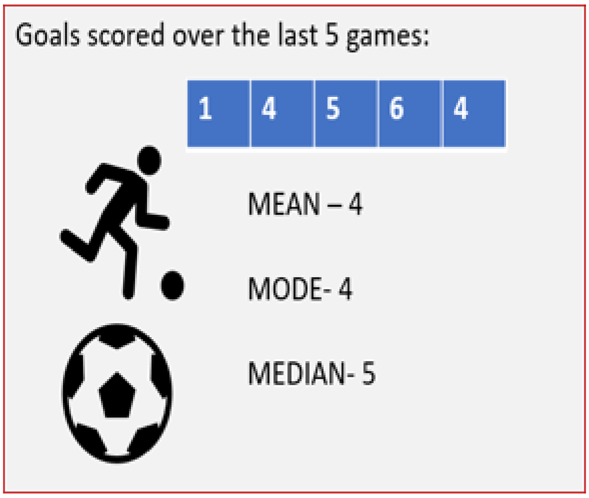
Statistical sampling

- The entire set of raw data that you may have available for a test or experiment is known as the population.
- You cannot necessarily measure the patterns and trends across the entire population.
- Take a sample, or portion of the population, perform some computations.

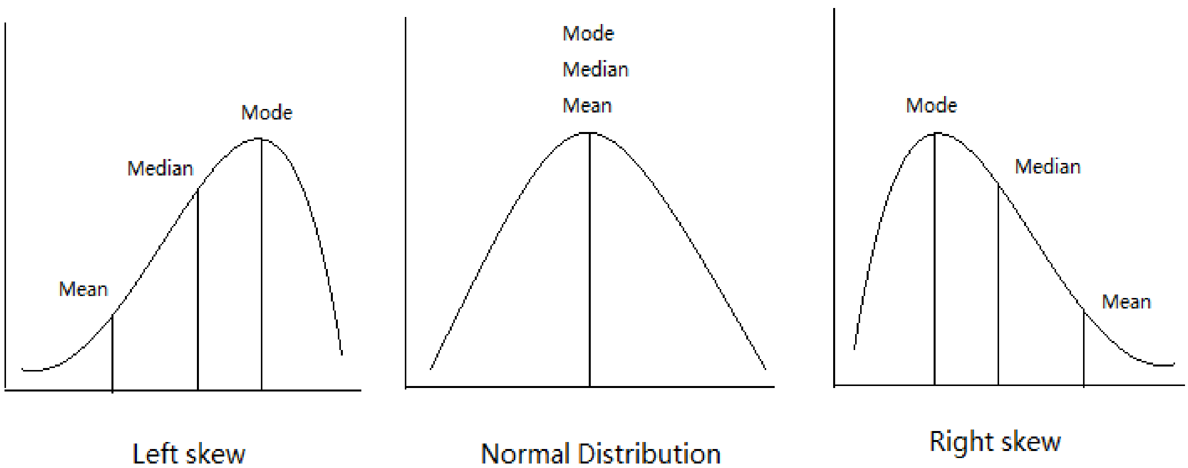


Descriptive statistics

- Helps us to describe the data and enables us to understand the underlying characteristics.
- Mean — the central value, commonly called the average.
- Median — the middle value if we ordered the data from low to high and divided it exactly in half.
- Mode – the value which occurs most often.



Distributions



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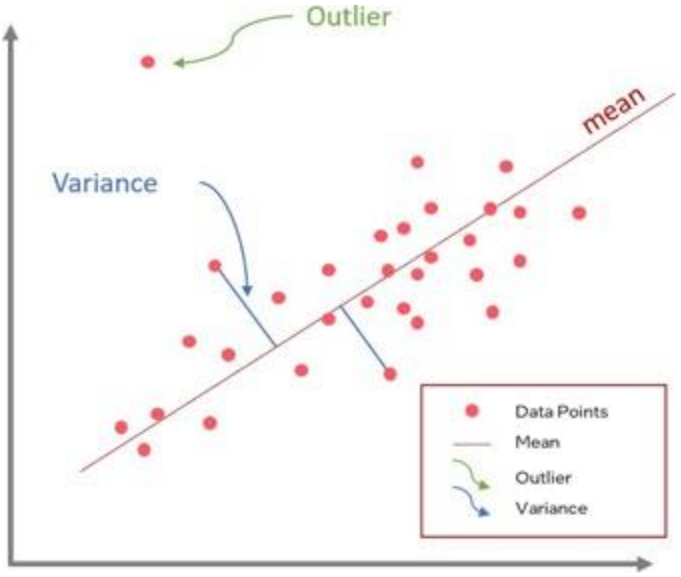
- These are charts or graphs that display the frequency of each value that appears in a data set.
- Some distributions contain some numbers much larger than others, causing the distributions to become skewed.
- Normal distribution – A distribution that is symmetrical in shape with most values around the central peak.

Probability

- In simple terms, is the likelihood of an event occurring.
- An event is the outcome of an experiment.
- Events can be either independent or dependent.

Variance

- Variance measures how far each value in the data set is from the mean (a measurement of the spread of numbers in a data set)
- Standard deviation – a calculation that gives a value to represent how widely distributed the values are.
- Outlier – A data point that lies at an abnormal distance from other values.



Activity: Mind map making

Purpose: Reinforce the need of Descriptive statistics

Say: Use any mind map tool offline/online to Craft a mind map in detailing the various contexts where mean, mode, and median are applied in real life.

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MS Excel for Statistical Data

Activity: Excel for Statistical Analysis

Purpose: To use Excel tool for Simple linear regression of cars speed and distance

Say: Download the template and follow the instructions

Activity template-MS Excel for Statistical Analysis: **https://bit.ly/43SIq6K**
Long Link: https://docs.google.com/spreadsheets/d/1f5G-JXyP7EV2fy1hax47YVaH5gyq8KZy/edit?usp=drive_link&ouid=109928090180926267402&rtpof=true&sd=true

Guidelines to Activity:

Use the Activity template Link to access the file:_________

Step 1: Get the Add-in

- Go to File
- Click on Options
- Click on Add-ins
    - Click on Analysis Tool Pak under Inactive Application Add-ins
    - In the bottom, you will find Manage-> Select Excel Add-ins-> Click on Go
    - Click Ok
- Add-in pop-up window will appear
    - Check on Analysis Tool Pak (Put a tick)
    - Click on Ok

- Once all the steps have been performed successfully, the Data Analysis option will appear inside Data Menu

Step 2: View the Speed Vs Distance Data available in the excel sheet

- Identify the Independent and Dependent Features – X and Y

Step 3: Visualize the Data using Scatter Plot

- Select both columns- Speed and Distance

- Go to Insert -> Charts -> Scatter
- Scatter Plot will appear
- Go to Chart Elements -> Edit Chart Title -> Distance Versus Speed

Step 4: Plot the Regression Line

- Click on the Scatter Plot
- Click on Chart Design
    - Add Chart Element
- Go to Trendline
- Click on More Trendline Options
- Click on Linear
- Put a tick on
- Display Equation on chart
- Display R-squared value on chart

Step 5: Verify the Linear Regression equation coefficients

- Click on Data
- Click on Data Analysis
    - Data Analysis option window will pop-up
    - Choose Regression
    - Click on Ok
    - Regression window will appear
- Select Y Range -> Select the Distance column data along with columnname Distance
- Select X Range -> Select the Speed column data along with columnname Speed
- Put a tick on the check box for Labels as we have selected the columndata along with column names
- Select output range and click on any cell where you want the output tobe displayed
- Summary output for regression statistics will appear on the sheet

Step 6: Find the distance using the generated linear equation as per y = mx + c

- For the value of Speed(X) = 6, What will be the distance covered by thevehicle?

Orange Data Mining

What is Orange Data Mining (ODM)?

- A machine learning tool for data analysis through Python and visual programming
- We can perform operations on data through simple drag and drop steps.
- Orange enables you to visualize data and perform data mining and machine learning.
- You can use this tool without writing a single code.
- This platform can be used for analysis, is relatively easy, and has beautiful visuals.

Orange Data Mining:

Orange is an open-source data mining and machine learning software suite designed for data analysis, visualization, and exploration. It offers a graphical user interface (GUI) that allows users tointeractively build data analysis workflows using various components called widgets. Each widget serves a specific purpose in the data analysis process. Here's a brief description of some commonly used Orange data mining widgets.

Data Loading Widgets: These widgets help you bring your data into Orange from files or online sources.

File: Allows you to load data from files in various formats such as CSV, Excel, and SQL.URL:

Loads data from a URL.

Data Table: Displays loaded data in a tabular format.

Data Exploration Widgets: These widgets allow you to look at your data in different ways, like scatter plots or histograms, to see patterns or trends.
Scatter Plot: Visualizes the relationship between two variables in the data.

Data Table: Allows for manual inspection and exploration of data.

Distributions: Displays histograms and other statistical distributions of variables.

Preprocessing Widgets: These widgets help you clean up your data, like filling in missing values or making sure all your data is on the same scale.
Impute: Handles missing values in the dataset. Normalize: Normalizes the data to a common scale.
Select Columns: Allows you to select specific columns from the dataset.

* Images shown here are the property of individual organisations and are used here for reference purpose only.

Feature Selection Widgets: These widgets help you choose which parts of your data are most important for your analysis.

Select Columns: Allows you to choose relevant columns/features from the dataset.

Select Best Features: Automatically selects the best features based on certain criteria like mutual information or correlation.

Modelling Widgets: These widgets build models from your data, like decision trees or clustering algorithms, to help you understand it better.

Classification Tree: Constructs a decision tree classifier.

k-Means: Performs k-means clustering on the data.

Support Vector Machine: Trains a support vector machine classifier.

Logistic Regression: Constructs a logistic regression model.

Evaluation Widgets: These widgets help you see how well your models are performing, so you can make adjustments if needed.

Test & Score: Evaluates the performance of a predictive model on a test dataset.

Cross Validation: Performs cross-validation to assess model performance.

ROC Curve: Plots the receiver operating characteristic curve for binary classifiers.

Visualization Widgets: These widgets help you turn your data into visual representations, like charts or graphs, to make it easier to understand.

Bar Chart: Displays data in a bar chart format.

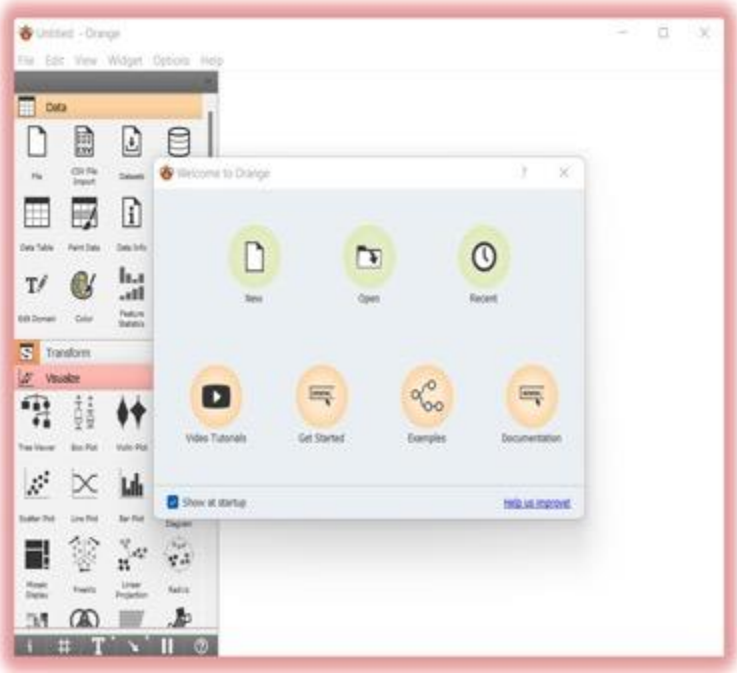
Heat Map: Visualizes data using a heatmap.

Scatter Plot: Visualizes the relationship between two variables.

These are just a few examples of the wide range of widgets available in Orange. Users can combine these widgets to create custom workflows for tasks such as data preprocessing, modelling, evaluation, and visualization, making it a powerful tool for data analysis and mining tasks.

Getting started with Orange Data Mining

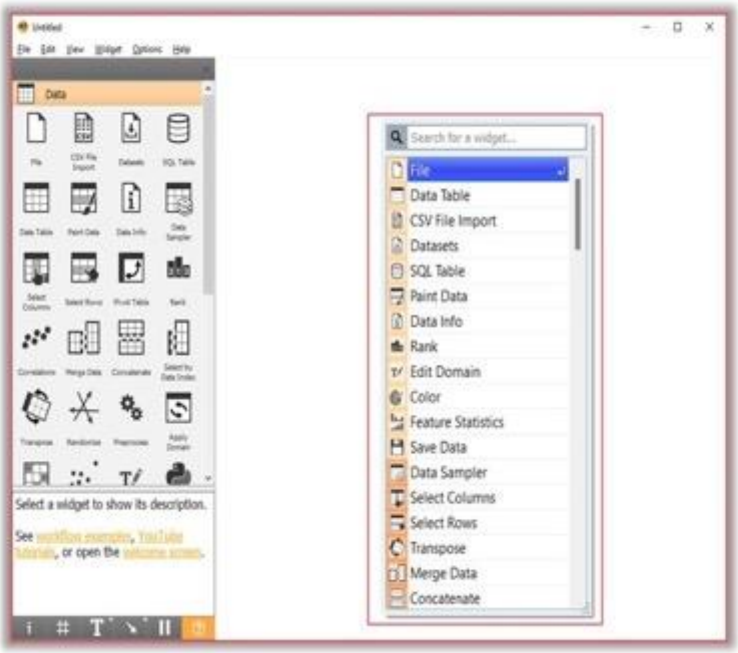
- Open the software, and this is what you will see!



When you open Orange you will see a welcome screen. We can start a new project here or open a project that has already been made. We can explore.

Exploring the Canvas

Right click on the canvas to get this prompt



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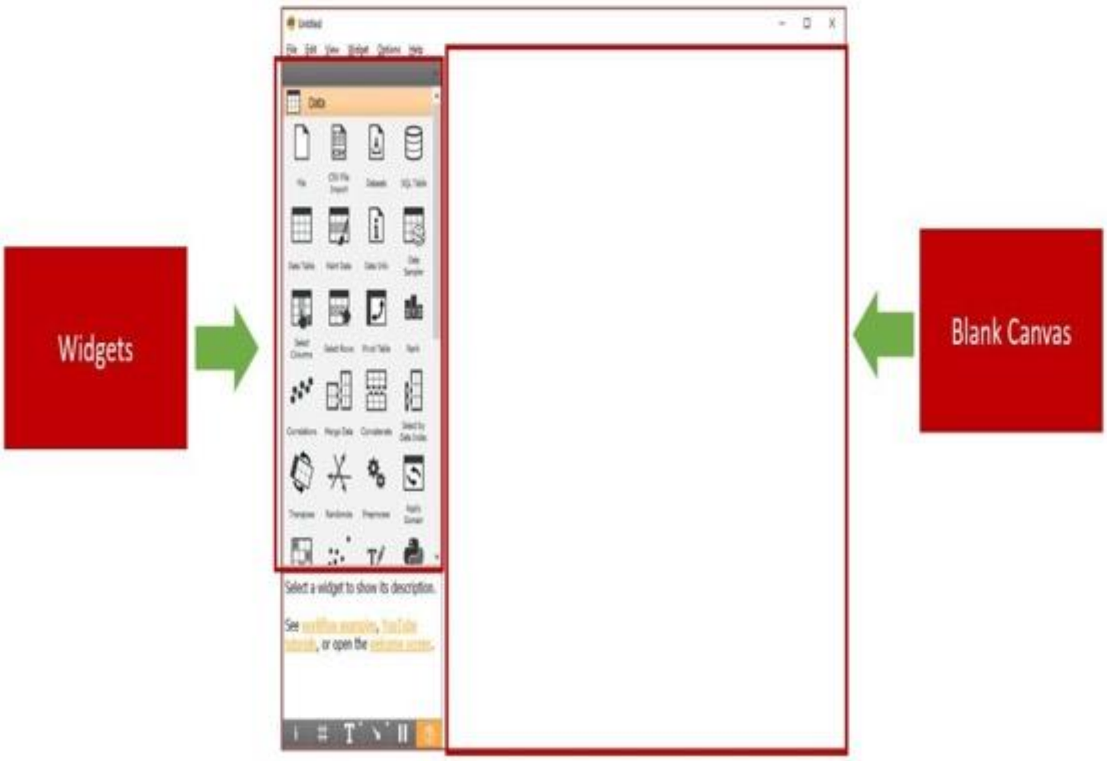
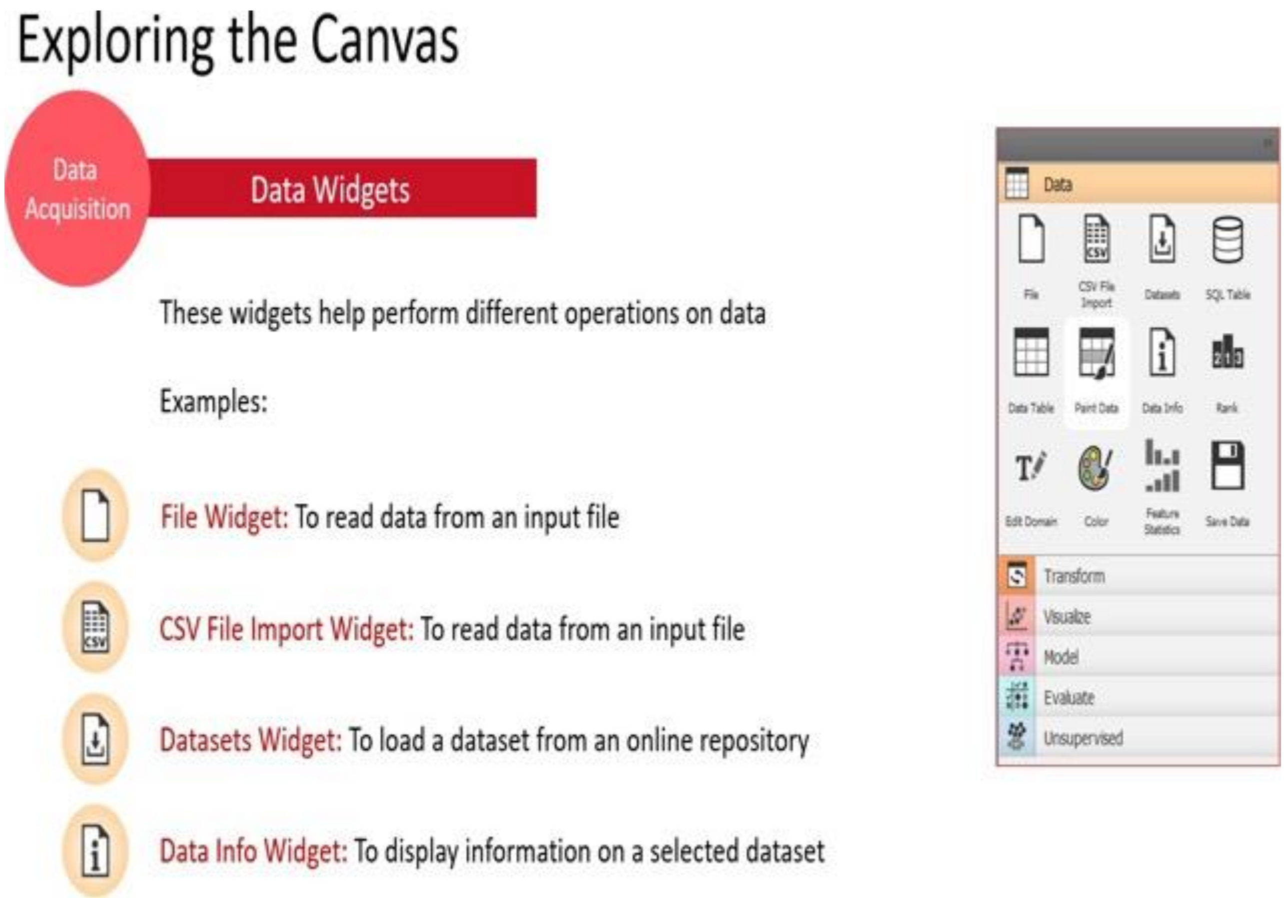


Orange comes with a lot of widgets to work with.
File widget, lets us load a file into our Workspace/canvas.



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Data Table widget, lets us observe what the data in the file looks like. Let's drag these two into our blank canvas (can also be done by clicking the widget)

Exploring the Canvas

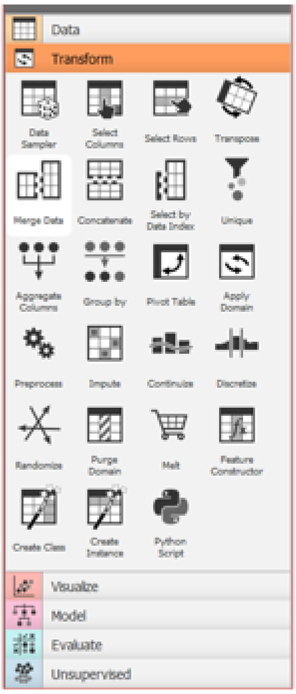
Data Exploration

Transform Widgets

These widgets help perform different operations on data

Examples:

- Data Sampler Widget: To create a subset of data points
- Select Columns Widget: To manually select data attributes
- Impute Widget: To replace unknown values in the data
- Discretize Widget: To discretize continuous attributes



Exploring the Canvas

Data Exploration

Visualize Widgets

These widgets help us visualize the data in different ways

For Example:

- Tree Viewer Widget: A visualization of classification and regression trees
- Box Plot Widget: shows the distributions of attribute values
- Scatter Plot Widget: The data is displayed as a collection of points
- Bar Plot Widget: Visualizes comparisons among categories

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Exploring the Canvas

Modeling

Model Widgets

These widgets help apply various models to the input data and use in our projects

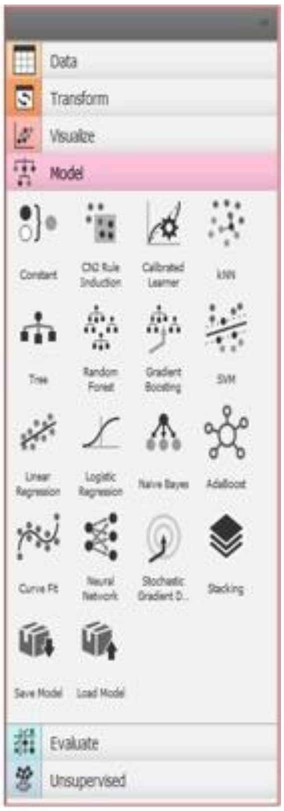
For Example:

Tree Widget: A simple algorithm that splits the data into nodes by class purity

Random Forest Widget: An ensemble learning method used for classification, regression, and other tasks

Linear Regression Widget: A linear regression algorithm

Logistic Regression Widget: The logistic regression classification algorithm



Exploring the Canvas

Modeling

Unsupervised Widgets

These widgets help us apply unsupervised learning models to our data and visualize it

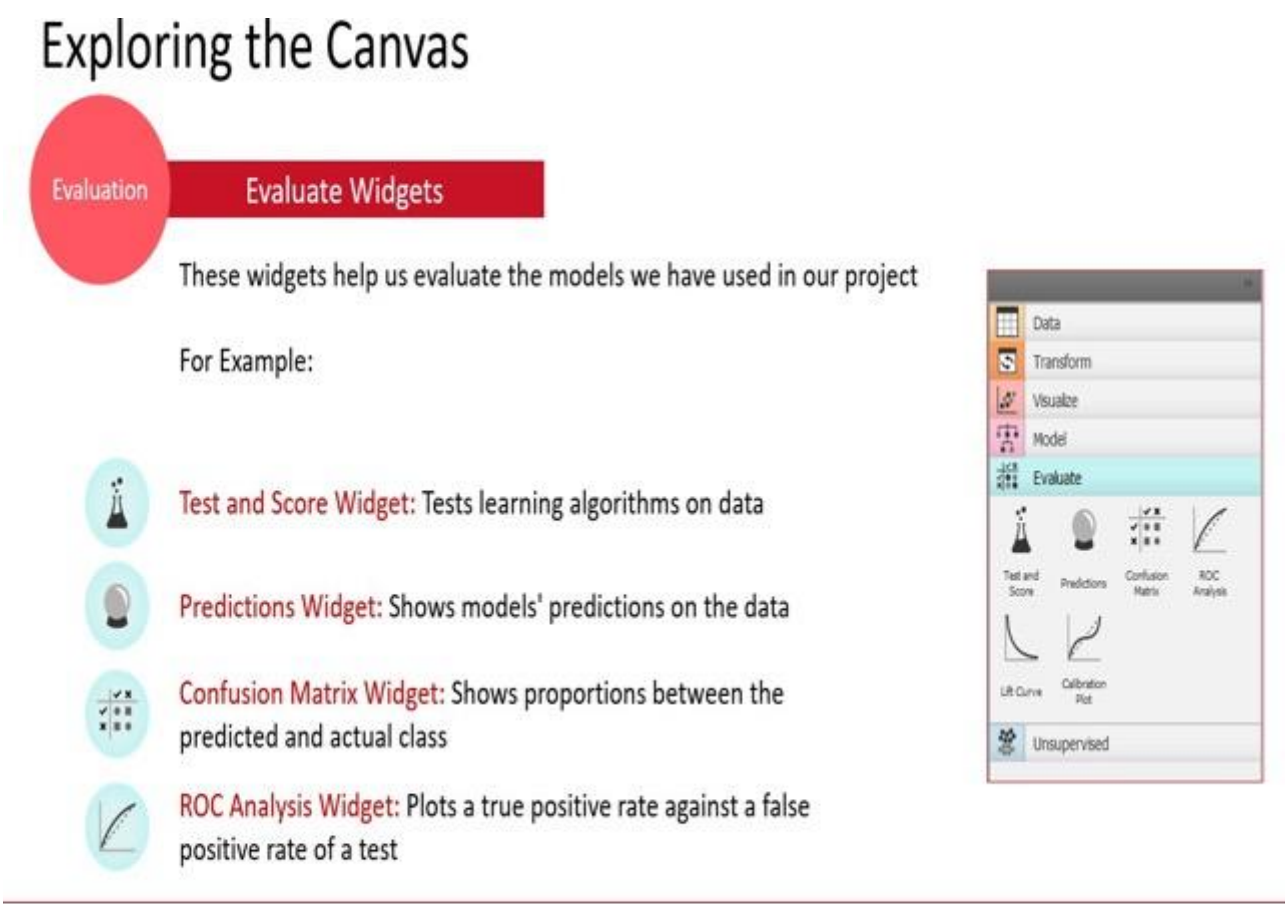
For Example:

Distance Matrix Widget: Visualizes distance measures in a distance matrix

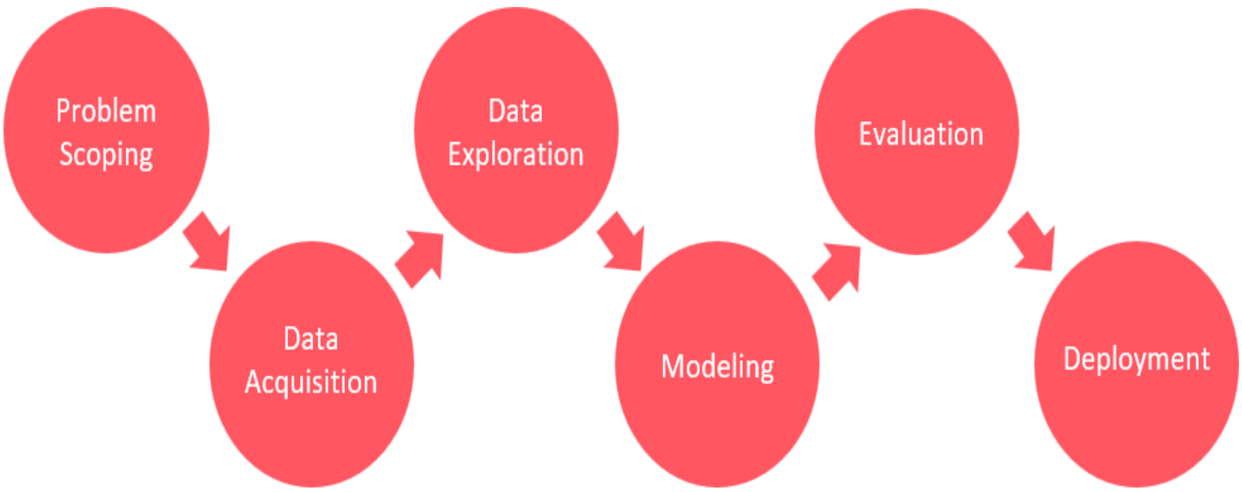
t-SNE Widget: Two-dimensional data projection with t-SNE

Correlations Widget: Compute all pairwise attribute correlations

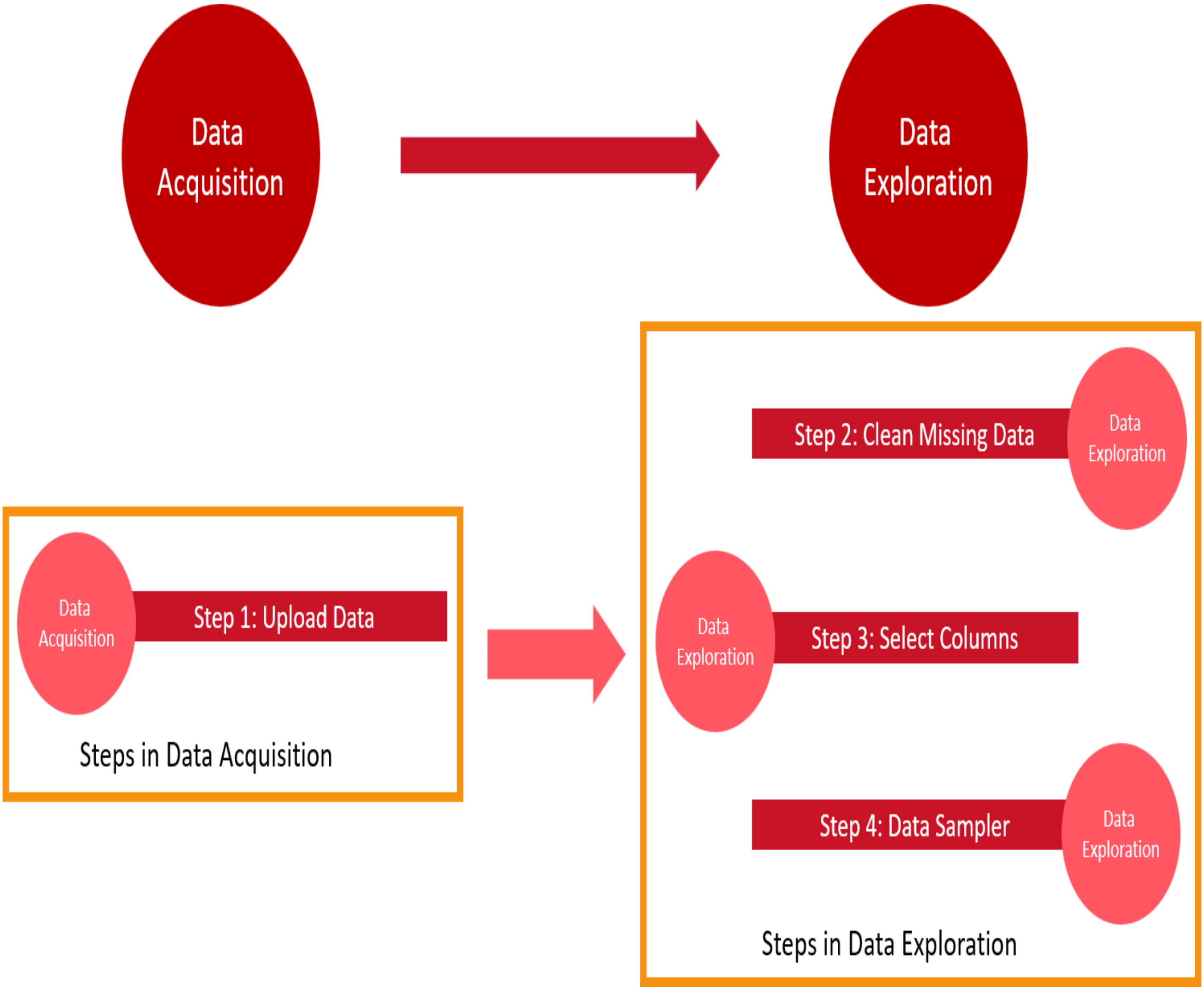
k-Means Widget: Groups items using the k-Means clustering algorithm

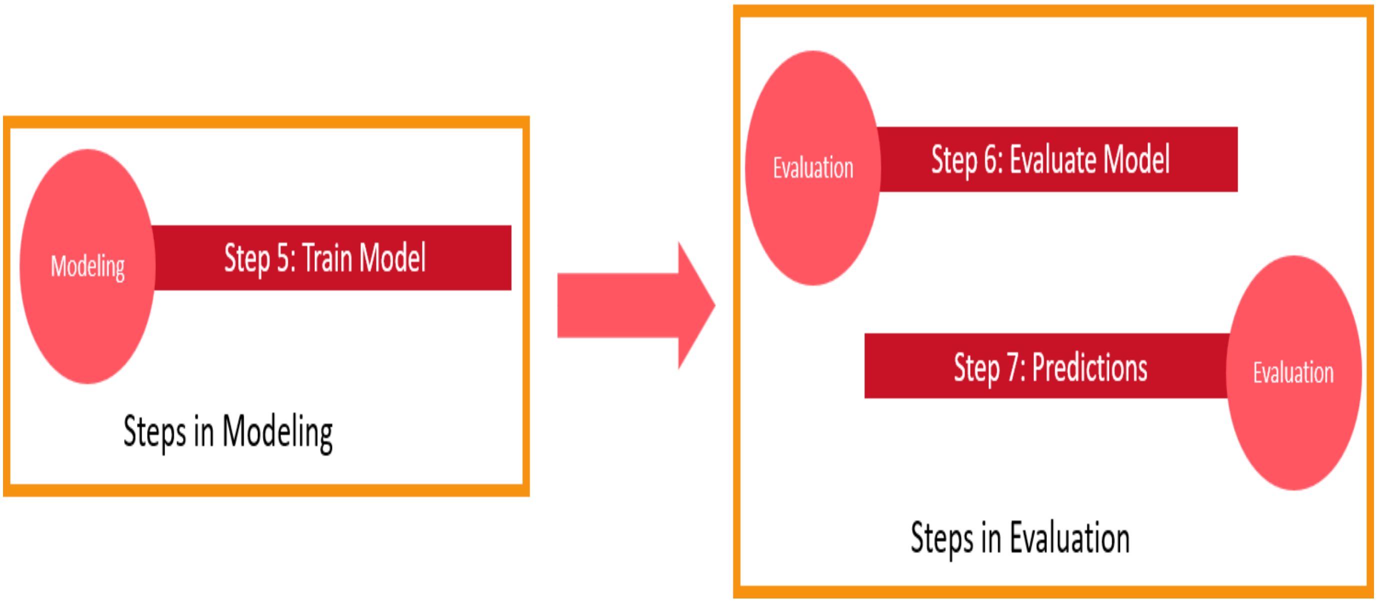


Recall the AI Project Cycle



Let’s understand how we can apply the AI Project Cycle in Orange Data Mining.





Thus, all the stages of AI project cycle can be mapped in Orange data mining tool.

CASE STUDY:

Palmer Penguins: Palmer Penguins are a species of penguin found in the Antarctic Peninsula region. Studies about penguins, including Palmer Penguins, often focus on their behaviour, habitat, population dynamics, and the effects of climate change on their ecosystems. Palmer Penguins dataset can be accessed from https://www.kaggle.com/code/parulpandey/penguin-dataset-the-new-iris/data

Student Portfolio - Homework

Palmer Penguins Model Index

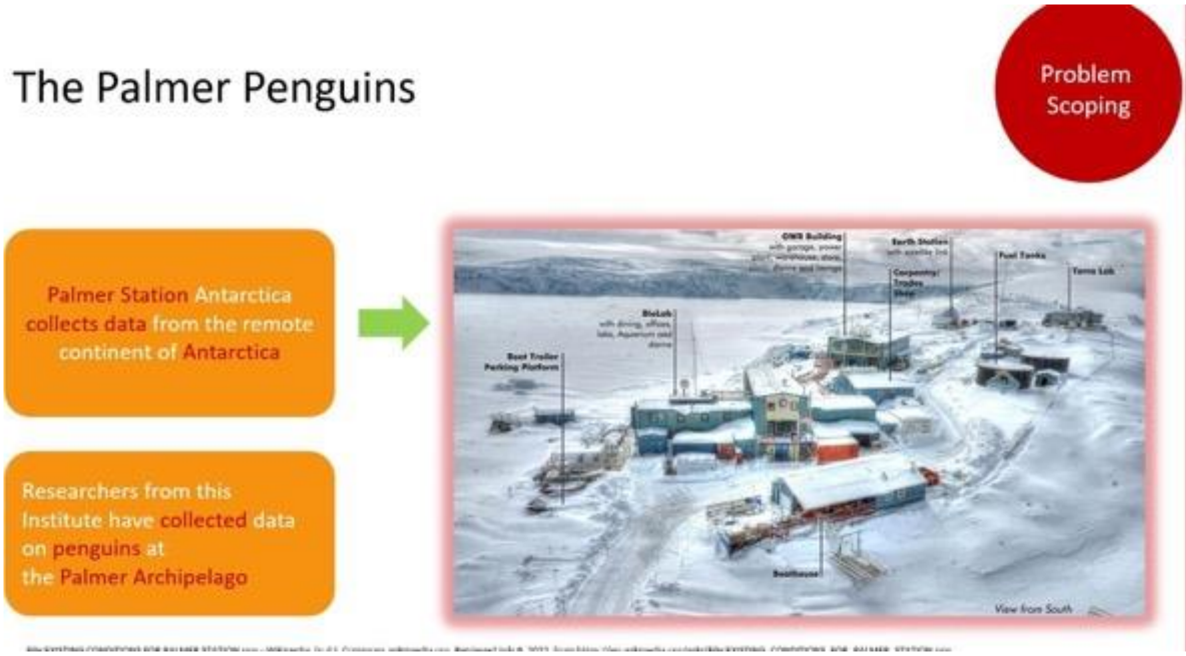
- Problem Scoping
- Data Acquisition (Step 1 Upload Dataset)
- Data Exploration (Step 2 Clean missing data)
- Data Exploration (Step 3 Select Target Label)
- Data Exploration (Step 4 Data Sampler)
- Modelling (Step 5 Train Model)
- Evaluation (Step 6 Evaluate model)
- Prediction (Step 7 Predictions)

The above image shows the index of this case study.

Scan the QR code for all files of PALMER PENGUINS.

Short link: https://bit.ly/4atlCN7

Long Link : https://drive.google.com/drive/folders/1fmcRVb-ilTyUhmUv4DWT1BFsaCoQ2BmF?usp=drive_link



Problem scoping

Our	Research community	Who
has a problem that	it is difficult to identify the species of some Palmer Penguins	What
when / while	collecting data from the remote continent of Antarctica	Where
An ideal solution would	predict the species of Palmer Penguins from the collected data	Why

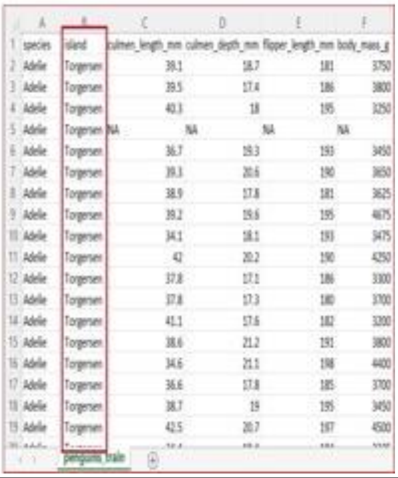
The researchers want to predict the species of Palmer Penguins based on the collected data.Can you help them? How?

Let us look at the dataset!
Look out for any differences in the physical features of the penguins!

The three species of penguins

Island feature

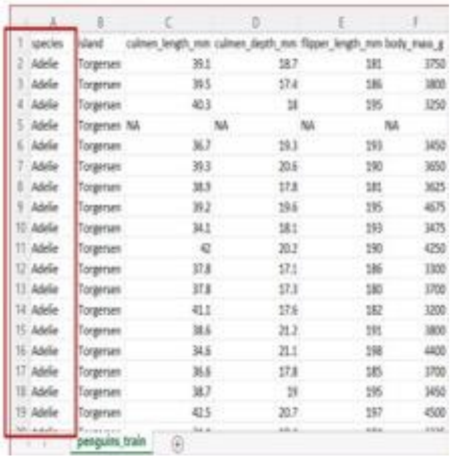
Island feature



	A	B	C	D	E	F
1	species	island	culmen_length_mm	culmen_depth_mm	flipper_length_mm	body_mass_g
2	Adelie	Torgersen	39.1	18.7	181	3750
3	Adelie	Torgersen	39.5	17.4	186	3800
4	Adelie	Torgersen	40.3	18	195	3250
5	Adelie	Torgersen	NA	NA	NA	NA
6	Adelie	Torgersen	36.7	19.3	193	3450
7	Adelie	Torgersen	39.3	20.6	190	3650
8	Adelie	Torgersen	38.9	17.8	181	3625
9	Adelie	Torgersen	39.2	19.6	195	4675
10	Adelie	Torgersen	34.1	18.1	193	3475
11	Adelie	Torgersen	42	20.2	190	4250
12	Adelie	Torgersen	37.8	17.1	186	3300
13	Adelie	Torgersen	37.8	17.3	180	3700
14	Adelie	Torgersen	41.1	17.6	182	3200
15	Adelie	Torgersen	38.6	21.2	191	3800
16	Adelie	Torgersen	34.6	21.1	198	4400
17	Adelie	Torgersen	36.6	17.8	185	3700
18	Adelie	Torgersen	38.7	19	195	3450
19	Adelie	Torgersen	42.5	20.7	197	4500

penguins_train

Species feature



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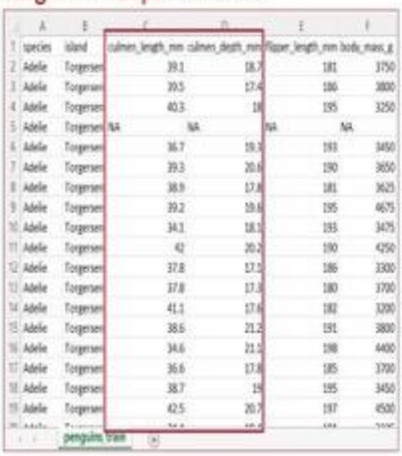
17	Adelie	Torgersen	36.6	17.8	185	3700
18	Adelie	Torgersen	38.7	19	195	3450
19	Adelie	Torgersen	42.5	20.7	197	4500

penguins_train

Culmen length and depth features



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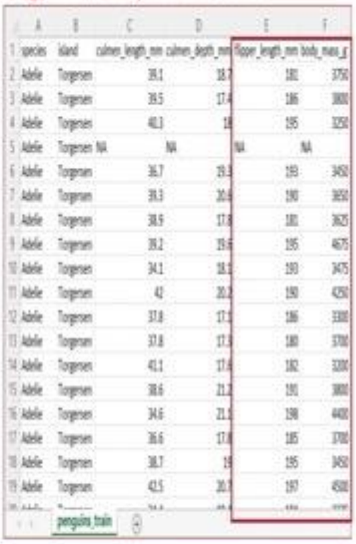
19	Adelie	Torgersen	42.5	20.7	197	4500

penguins_train

Flipper Length and Body Mass feature



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2	Adelie	Torgersen	39.1	18.7	181	3750
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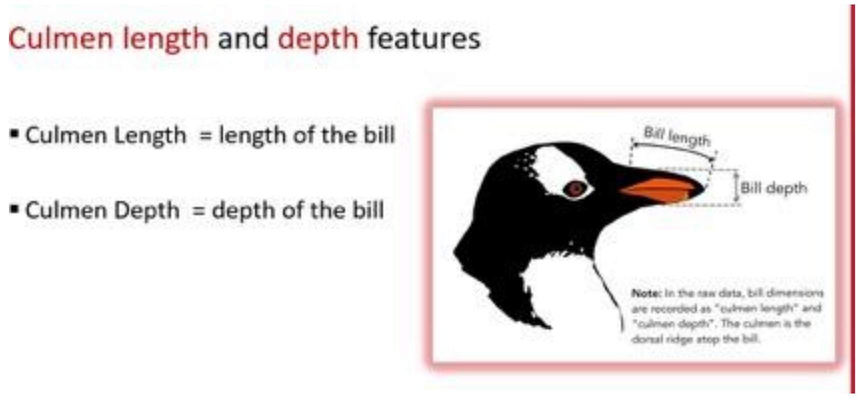
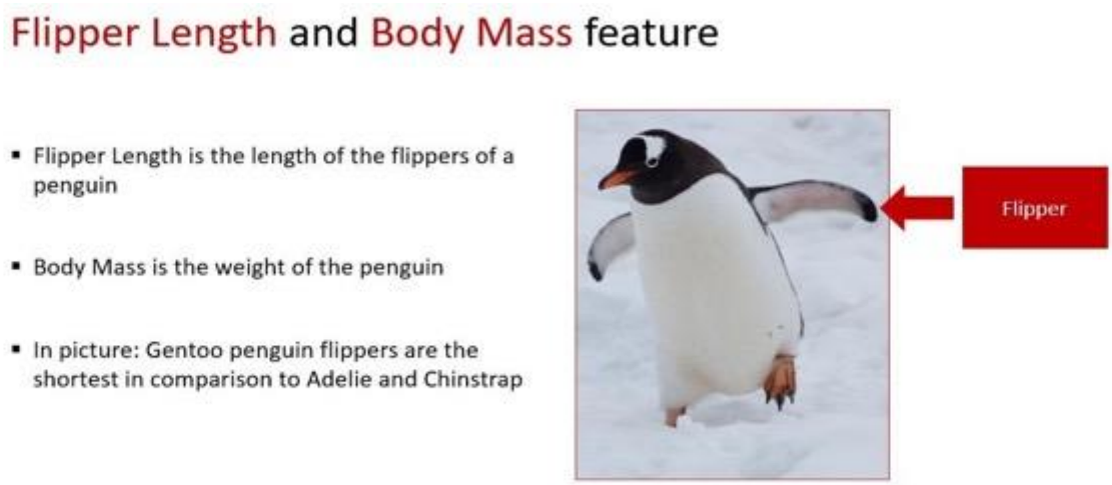
penguins_train





Let’s teach an AI model to predict the penguin species in Orange Data Mining now!

Based on features like their sizes, how long their beaks are, and the colors of their feathers. Now, youwant to figure out which species of penguin is which. ODM is a smart tool that can analyze all the penguin data you collected and learn to recognize patterns.

The link for steps creating the AI model can be accessed in the file

Orange datamining.pdf

Reflection

Are there any limitations of using No-Code AI tools? Share your thoughts with the class?

Test Yourself:

1.What type of tool is Orange?

a) Open-source b) Closed-source

c) Paid software d) Hardware

2. Which of the following tasks can be performed using Orange?

a) Classification b) Regression

c) Clustering d) All of the above

3. What type of data can be imported into Orange?

a) CSV b) Excel

c) SQL databases d) All of the above

4. What does the Data Table widget in Orange primarily facilitate?

a) Loading and viewing datasets b) Performing clustering analysis

c) Running machine learning algorithms d) Evaluating

5. Which component in Orange enables users to evaluate the performance of machine learning models?

a) Test & Score b) Data Table

c) Data Projection d) Data Exploration

Reflection Time:

1. Define No-Code and Low-Code AI.
2. Identify the differences between Code and No-Code AI concerning Statistical Data.
3. Relate AI project stages to the stages of No-Code AI projects.
4. Orange Data Mining is a machine learning tool for data analysis where we can perform operations through simple drag-and-drop steps.
5. Descriptive statistics – Mean, Median, Mode – help us to describe the data and its underlying characteristics.
6. No code tools for AI projects.

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